

RIGIDITY AND UNIVERSAL CONTRACTIONS OF HARD-SPHERE KINETIC COTANGENTS

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ABSTRACT. Each phase is represented by a genuine restricted finite-volume law with its own local analytic pressure. Coexistence allows Lee–Yang pinching and is treated by a microscopic subexponential mixture theorem, which directly yields the minimum of phase rates. Phase-wise face recovery, projective passage, thin-shell conditioning, and exact finite-centered likelihoods remain a standalone positive theory.

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1. ROUND-EIGHT RECONSTRUCTION: PHASE-RESTRICTED MICROSCOPIC LAWS AND STRATIFIED MIXTURE LDPs

Coexistence is handled by genuine phase-restricted finite-volume objects and an exponential mixture theorem. Lee–Yang zeros may pinch the coexistence set; there is no common zero-free tube there. The minimum of phase rates is proved from the microscopic mixture and is not inferred from the conjugate of a maximum pressure.

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1.1. Phase-restricted partition functions. Let $j = 1, \dots, J$ label metastable boundary conditions or spectral phase projectors. Define finite positive phase laws $\mathbb{P}_{\varepsilon,j}$ and restricted partition functions

$$Z_{\varepsilon,j}(\theta) = \mathbb{E}_{\varepsilon,j} e^{\mu_\varepsilon \theta \cdot X_\varepsilon}.$$

On the interior $\mathcal{U}_j$ of phase j , assume the restricted transfer operator has one isolated eigenvalue. Its logarithm is analytic on local complex tubes compactly contained in $\mathcal{U}_j$.

Lemma 1.1 (Local phase atlas only). *For every compact $K \Subset \mathcal{U}_j$, the functions $\mu_\varepsilon^{-1} \log Z_{\varepsilon,j}$ converge analytically on a fixed complex neighborhood of K . No such neighborhood is asserted across a coexistence point. Zeros of the total partition function are allowed to approach coexistence at order μ_ε^{-1} .*

Proof. The isolated phase eigenvalue and its Riesz projector persist on a compact subdomain of one phase, giving a nonvanishing restricted leading term and analytic perturbation. At coexistence two restricted exponentials have equal real part, so their sum can have nearby complex zeros; excluding that set from the phase tube is both necessary and consistent. $\square$

1.2. Phase-wise finite-dimensional LDPs. For a finite projection π_m , let $Q_{m,j}$ be the phase pressure and $I_{m,j} = Q_{m,j}^*$ on the interior mean image. Boundary points are recovered inside the same microscopic phase by normal concentration followed by a finite tangential tilt.

Theorem 1.2 (Phase-wise face LDP). *Each restricted law $\mathbb{P}_{\varepsilon,j} \circ \pi_m^{-1}$ satisfies a full good LDP with rate $I_{m,j}$ on its attainable compact support. For an exposed face F , the lower recovery is obtained by first choosing a large but finite normal source, passing $\varepsilon \rightarrow 0$, and then sending the normal source to infinity while applying the phase-specific tangential coefficient theorem.*

Proof. Interior points use the analytic phase saddle. Compact support gives the upper bound without an infinite source domain. For a face, exponential tightness in the normal coordinate makes the finite normal tilt concentrate in a small neighborhood of F . Conditional on that neighborhood, the tangential restricted transfer operator remains inside phase j and its local-limit coefficient gives the lower bound. A diagonal sequence in the normal source completes the recovery. $\square$

1.3. The microscopic mixture theorem. Assume the physical law has the decomposition

$$\mathbb{P}_\varepsilon = \sum_{j=1}^J w_{\varepsilon,j} \mathbb{P}_{\varepsilon,j} + R_\varepsilon,$$

where

$$\mu_\varepsilon^{-1} \log w_{\varepsilon,j} \rightarrow 0, \quad \|R_\varepsilon\| \leq e^{-c\mu_\varepsilon}.$$

Theorem 1.3 (Coexistence mixture LDP). *The physical finite-dimensional law satisfies a full LDP with rate*

$$I_m(x) = \min_j I_{m,j}(x).$$

Its pressure is $Q_m(\theta) = \max_j Q_{m,j}(\theta)$, but the rate identity is a consequence of the microscopic mixture, not the generally false formula $(\max_j Q_{m,j})^ = \min_j Q_{m,j}^*$.*

Proof. For a closed set, sum the phase upper bounds; the largest exponential term is governed by $\inf_x \min_j I_{m,j}(x)$. For an open set and a point x , choose a phase attaining the minimum and use its lower recovery from Theorem 1.2; the subexponential mixture weight does not change the rate. The signed remainder is exponentially negligible. The pressure maximum follows separately by the log-sum-exp identity. $\square$

1.4. Projective passage and topology upgrade. Let $I_j = \sup_m I_{m,j} \circ \pi_m$ and $I = \min_j I_j$.

Theorem 1.4 (Stratified projective LDP). *If every phase family is exponentially tight in the target weighted topology and every finite-rate point has a phase-wise recovery sequence exponentially concentrated in one compact rate sublevel, then the physical laws satisfy a full target-topology LDP with rate I .*

Proof. Apply the projective theorem phase by phase. Compact rate sublevels make the projective and target topologies agree on the recovery supports. For a target-open set, choose a projective neighborhood whose intersection with the selected compact sublevel lies inside it. The recovery law puts superexponentially small mass outside that compact, so its projective lower bound is also a target lower bound. Combine phases with Theorem 1.3. $\square$

1.5. Conditioning and exact local likelihoods.

Theorem 1.5 (Phase-stratified thin-shell conditioning). *Suppose each active phase has the A2 or B1 exact finite-source shell coefficient uniformly on its phase chart. Conditioning the microscopic mixture on the shell produces a new subexponential mixture of the conditioned phase laws. The conditional LDP has the minimum of their constrained rates.*

Proof. Apply the phase coefficient theorem to each numerator and denominator. Phases whose shell exponent is larger are exponentially suppressed; phases tying at the minimum retain subexponential weights. The conditioned law is therefore again of the form required by Theorem 1.3, with each phase rate restricted to the shell. $\square$

Theorem 1.6 (Exact finite-centered likelihood and contraction). *Within every active phase, the local likelihood ratio is centered at the exact finite-volume mean and has the Gaussian quadratic form supplied by the phase covariance. Under any continuous contraction, phase rates contract first and*

are then minimized over phases; this order agrees with direct contraction of the microscopic mixture.

Proof. Taylor expand the restricted finite log partition function at its exact saddle; the score has zero finite expectation and the Hessian is the exact covariance. For contraction, push forward each phase law and apply the ordinary contraction principle. Pushforward commutes with the finite mixture, so Theorem 1.3 gives the minimum of the phase-wise contracted rates. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states positive theorem closure in the declared local source, short-time, finite-horizon or Boltzmann–Grad regime specified in its theorems. Every load-bearing cross-paper input is typed in the round-eight dependency ledger. Clean compilation and automated structural checks are necessary verification gates; independent external journal review remains a separate process.

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